

Section 92 — Discrete Logarithms And Modular Roots

Invert exponentiation by exposing group structure

This lesson derives A–C. Problem D is editorial-only.

A. Primitive roots of a prime

The nonzero residues modulo prime p form a cyclic group of order $p-1$. A candidate g has full order exactly when

$$g^{(p-1)/q} \neq 1 \pmod p$$

for every distinct prime divisor q of $p-1$. If its order were proper, it would divide $(p-1)/q$ for some missing prime factor; conversely any proper divisor misses one such factor. Factor $p-1$, then test candidates upward to obtain the smallest root.

B. Baby-step–giant-step

Write an exponent as $x=i*m+j$, with m near $\sqrt{\text{modulus}}$. Store baby powers a^j . Walk giant values $b*(a^m)^{-i}$ until one matches. The equality is exactly $a^{(im+j)}=b$. Retaining the first baby index and minimizing candidates returns the smallest exponent.

Time and memory are both $O(\sqrt{m})$. This is a meet-in-the-middle algorithm: split one long unknown exponent into two short coordinates.

C. When the inverse does not exist

If $g=\gcd(a,m)>1$, every positive power of a is divisible by g . Therefore b must also be divisible by g ; otherwise no solution exists. Divide the modulus and target by g , record one forced exponent, and update the accumulated scale. Repeat until base and reduced modulus are coprime, then use BSGS on the adjusted target.

Check $x=0$ and the accumulated scale at every reduction. Those early solutions are precisely what a careless coprime-only derivation loses.

Recognition guide

First identify the algebraic domain: a prime-field multiplicative group, a smaller subgroup, or a non-group monoid containing nonunits. Group algorithms require inverses; a gcd reduction is not an optimization but the bridge that makes those inverses legal.

A BSGS equation without hand-waving

Choose $m = \text{ceil}(\text{sqrt}(\text{order_bound}))$ and write $x = i * m + j$, with $0 \leq j < m$. Then

$$\begin{aligned} a^x &= b \\ a^j &= b * (a^m)^{-i}. \end{aligned}$$

The left side depends only on j , so store it. The right side changes by one fixed inverse giant step. If multiple pairs collide, compute their actual exponents and keep the smallest legal one; the first hash-table collision is not automatically the smallest solution unless iteration and stored indices were designed to guarantee that.

Tonelli–Shanks as shrinking two-adic order

For odd prime p , write $p-1 = q * 2^s$ with odd q . A quadratic non-residue supplies an element of exact two-power order, while powers of the target give an initial candidate root and a residual error. Each iteration finds the first squaring of that error equal to one, multiplies by a suitable power of the non-residue, and strictly reduces the error's two-adic order. That decreasing order is the termination measure.

Before any of this, Euler's criterion distinguishes zero, a non-residue, and a nonzero residue. Special-case $p=2$; when $p \bmod 4 = 3$, the direct exponent $a^{((p+1)/4)}$ is a useful simplification.

Testing the algebraic layers

For small moduli, enumerate exponents to obtain the smallest discrete log and enumerate residues to obtain all square roots. Include $a=0$, $b=1$, nonunits, no-solution cases, and logs equal to zero. Verify a returned answer by modular exponentiation, but remember that this alone does not prove minimality; the small oracle should compare the exponent itself.

Exercises

A–C cover group generators, meet-in-the-middle logarithms, and non-coprime reduction. D asks for square roots through the two-adic structure of $p-1$.